AI & Machine Learning Internship – Task 2

[Feature Engineering, Model Optimization & Performance Comparison]

Objective

The objective of this project was to improve the house price prediction model developed in Task 1 by applying feature scaling, training multiple regression models, and comparing their performance. The California Housing Dataset was used to predict house prices based on housing-related features.

Methodology

The California Housing Dataset was loaded using the scikit-learn library. The dataset contains features such as median income, house age, average rooms, average bedrooms, population, occupancy, latitude, and longitude.

To improve model performance, feature scaling was performed using StandardScaler. Scaling ensures that all features are on a similar numerical scale, allowing machine learning algorithms to learn more effectively.

The dataset was divided into training and testing sets using an 80:20 ratio. Three regression models were trained and evaluated:

1. Linear Regression
2. Ridge Regression
3. Decision Tree Regressor

Model performance was measured using Root Mean Squared Error (RMSE) and R^2 Score.

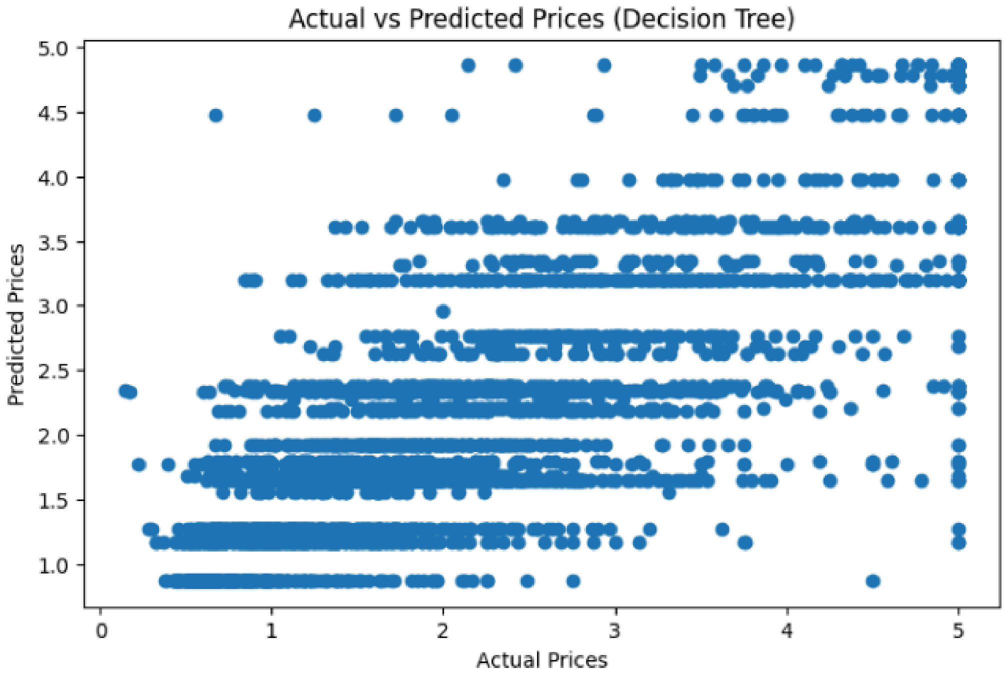
Results

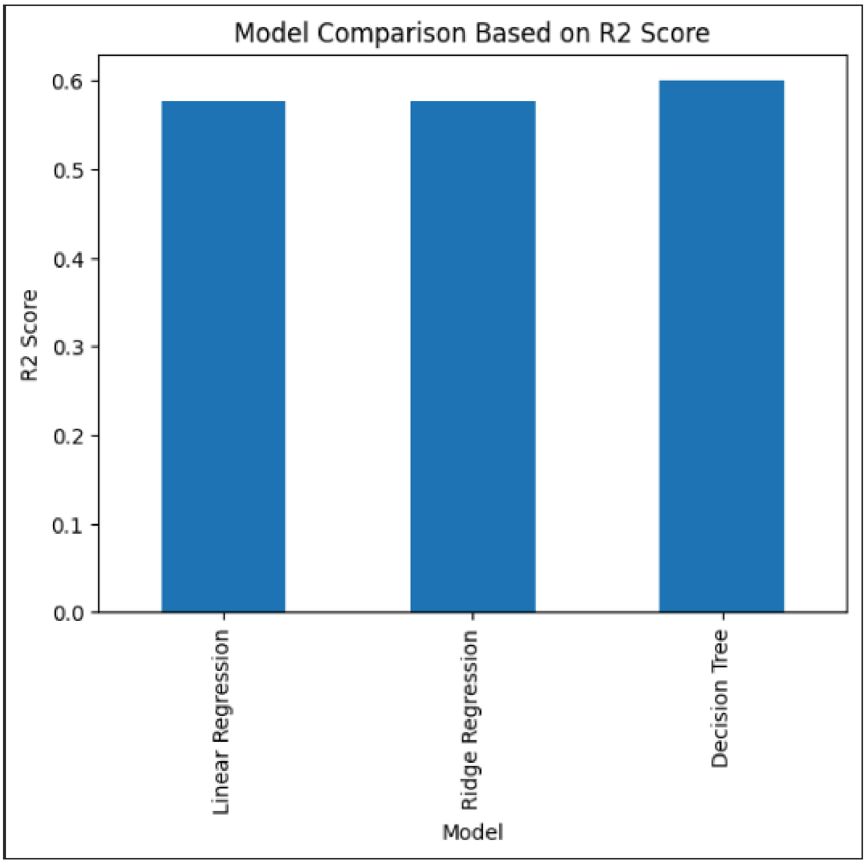
After training and testing the models, their performance metrics were compared.

	Model	RMSE	R2 Score
0	Linear Regression	0.745581	0.575788
1	Ridge Regression	0.745554	0.575819
2	Decision Tree	0.724234	0.599732

A comparison chart was also generated to visualize model performance. The model with the highest R^2 Score and lowest RMSE was selected as the best-performing model.

	Model	RMSE	R2 Score
0	Decision Tree	0.724234	0.599732
1	Ridge Regression	0.745554	0.575819
2	Linear Regression	0.745581	0.575788





Conclusion

This project demonstrated the importance of feature engineering and model comparison in machine learning. Feature scaling helped standardize the input features, while training multiple models allowed objective performance evaluation.

Among the tested models, the best-performing model achieved the highest predictive accuracy based on the evaluation metrics. The results showed that comparing different algorithms is an important step in building reliable machine learning solutions.

This task provided practical experience in preprocessing, model optimization, performance evaluation, and model selection using real-world housing data.